

Internal Assessment

Physics – Higher Level

Exploring the impact of windage on projectile motion

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1.Introduction

In a competition, we had a fun game of building a model cannon, aiming for maximum range. While doing this, I noticed that the paper ball went different distances for different cannons, even though the cannons were almost identical. However, there was one difference, the girth of the barrel. This observation intrigued me and hence inspired me to explore in depth about this topic in my Physics IA. This investigation delves into how windage—the gap between the projectile and the barrel—affects the flight performance of a projectile. Making the propulsion system from scratch, I varied the barrel diameter systematically and recorded the projectile's range, and used video analysis to capture the flight time and impact point of the projectile. The results revealed a fascinating trend which was not what I expected.

2.Research Question

How does the distance between the projectile and the barrel (windage) (0.119, 0.269, 0.419, 0.569, 0.719, 0.869 cm) affect the range of a projectile?

3.Theory and Hypothesis

3.1. Background Theory

One of the fundamental concepts of classical physics is Projectile Motion, which describes the motion of a moving body via a combination of the horizontal and vertical components of its motion through a medium. The vertical component is influenced by gravitational acceleration acting downwards (Chudinov), whose kinematic equation can be described as

$$y = v_{oy} \cdot t - \frac{1}{2}gt^2$$

where v_{oy} is the initial vertical velocity and g is the gravitational acceleration. However, this is not the area of focus for this particular investigation (“3.4: Projectile Motion”).

The horizontal component is characterized by constant velocity since there are no horizontal forces acting on the projectile (assuming negligible air resistance). The horizontal distance x can be expressed as

$$x = v_{ox} \cdot t$$

Where (3.4: Projectile Motion”):

- v_{ox} - initial horizontal velocity
- t - time of flight.

Range is the distance covered by a projectile in the x-direction (Admin). Since this investigation was primarily about range, it will focus mainly on the x-components.

Windage is the space between the projectile and the surface of the barrel (“Merriam-Webster Dictionary”). This distance can significantly affect the projectile's launch dynamics.

Figure 1: Windage (Kobus)



As windage increases, the projectile may experience changes in its initial velocity and trajectory due to:

1. **Air resistance:** Larger windage allows more airflow around the projectile, increasing drag and reducing range.
2. **Stability and spin:** Windage affects aerodynamics too, and may cause wobbling or irregular spin, decreasing range.
3. **Energy dissipation:** Excessive windage may lead to collisions with the sides of the barrel causing energy loss and affecting range (Bernardo et al.).

3.2. Hypothesis

Based on these reasons, I hypothesis the following:

Null hypothesis: The range of the projectile will decrease as the windage increases.

Alternate hypothesis: The range of the projectile will increase as the windage increases.

4.Apparatus

Table 1: Apparatus List

Materials and Instruments	Uncertainty
PVC pipes with varying diameter	-
Soldering iron	-
Pencil	-
2 extension springs	-

Retort stands	-
Retort stand clamps	-
Ring Clamp	-
Marble	-
Measuring tape	$\pm 0.0005 \text{ m}$
Protractor	$\pm 0.5^\circ$
Plumb line with metal bob	-
Sand	-
Chart paper	-
Vernier calliper	$\pm 0.00005 \text{ m}$
Micrometre screw gauge	$\pm 0.000005 \text{ m}$
Smart phone with a video camera	$\pm 0.01 \text{ s}$

5.Variables

5.1. Independent variable

Table 2: Independent variable

Variable	Method	Range ($\pm 0.0005 \text{ m}$)
Windage – m	Using different diameter pipes as barrels	1.83, 2.13, 2.43, 2.73, 3.03, 3.33

5.2. Dependent variable

Table 3: Dependent variable

Variable	Method
Range (horizontal distance) – m	Using different diameter pipes as barrels for the same propulsion system

5.3. Controlled variables

Table 4: Controlled Variables

Variable	Rationale for control	Control Method
Launch Angle	The launch angle directly impacts the projectile's trajectory and range, making it essential to maintain a constant angle for reliable comparison	Use a plumb line to set the propulsion system to 45°, as this angle would provide a balance between range and height
Type of projectile	Variations in mass, shape, or surface texture of the projectile affect drag and inertia, thus requiring a uniform projectile for consistent results.	Use the same marble for all launch trials, ensuring it is not damaged

Propulsion system	The propulsion system must remain unchanged across trials to ensure consistency in the experiment.	Use a single, consistent propulsion system (a self-made extension-spring loaded propulsion system) throughout the experiment
Environmental conditions	External factors like wind, humidity, and temperature can alter aerodynamic behaviour, so controlling these variables is crucial for data integrity.	Conduct all trials indoors in a temperature-controlled physics laboratory
Measurement tools	Precision and accuracy of measurement tools are vital in maintaining the reliability of recorded data.	Use the same measuring instruments for measurement of variables in all trials to ensure consistency

6. Experimental Design

After research (Cannon) and adapting to the experiment's needs, the propulsion system was designed. Two parallel slits were made on opposite sides of a PVC pipe using a soldering iron to create a guiding tube for the pusher mechanism. A smaller diameter pipe, cut into a short piece, served as the pusher for the marble. Its diameter, smaller than both the marble and guiding tube, allowed the marble to rest securely when loaded and ensured free motion inside

the tube. A hole drilled into the pusher aligned with the slits, and a short pencil was threaded through, protruding from both slits to limit the pusher's movement.

The guiding tube, held at a 45-degree angle by two retort stand clamps (angle determination detailed in steps 4-7 in the methodology), had springs attached on either side of the pencil, with their opposite hooks secured to a ring clamp above the open end, as shown in Figure 2 to Figure 4

Figure 2: Barrel set-up

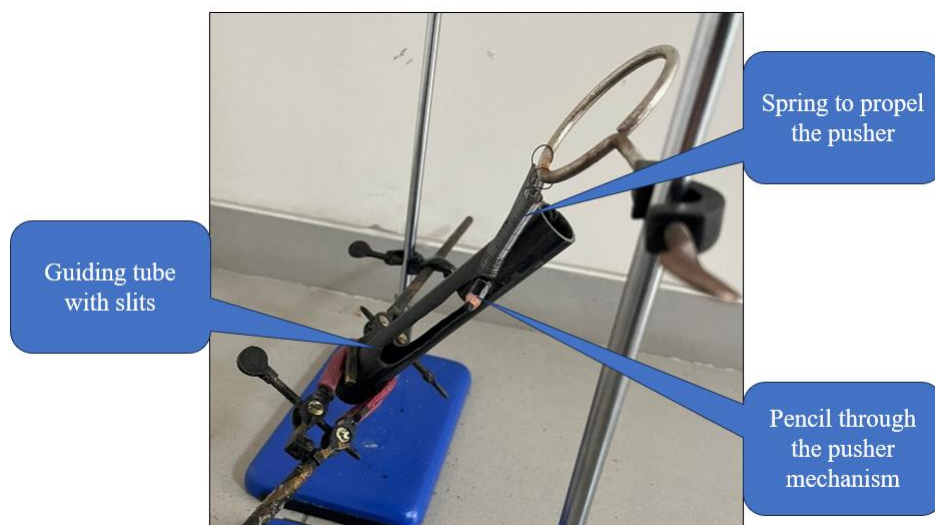
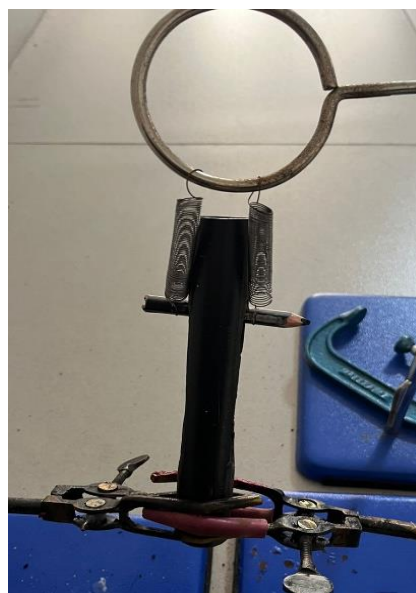


Figure 3: Top view



To measure the marble's range, a chart paper was laid on the expected impact zone. A smooth layer of sand was spread on the paper to cushion the marble's fall to prevent damage and forming a visible crater at the point of impact for accurate range measurement, as in Figure 4

Figure 4: Barrel set-up



The smartphone was attached to a retort stand using clamp and was placed such that the entire experimental area was visible in the camera, as in Figure 5

Figure 5: The entire set-up



7. Experimental Methodology

1. Set the propulsion system parallel to a wall and lay sand-covered chart paper in the expected impact area.
2. Extend the measuring tape from the body tube's closed end to beyond the expected impact zone, laying it parallel to the wall.
3. Attach a protractor parallel to the body tube using a clamp, positioning it upside down.
4. Use a plumb line to align the body tube at a 45-degree angle, then remove the protractor and clamp, as shown in Figure 6 below.
5. Attach the barrel of known diameter to the body tube with a retort stand and clamp.
6. Verify the barrel is also at the 45-degree angle by repeating steps 3-5.
7. Start video recording with a smartphone.
8. Load the marble into the barrel, resting it on the pusher.
9. Pull the pusher to the end of the slits and release it.
10. After the ball hits the sand, stop the video and measure the range from the crater to the body tube using measuring tape as shown below in Figure 6-7.
11. Smoothen the sand.
12. Repeat steps 7-11 for 10 trials per barrel and for each barrel diameter and take the average.

Figure 6: Launch angle adjustment

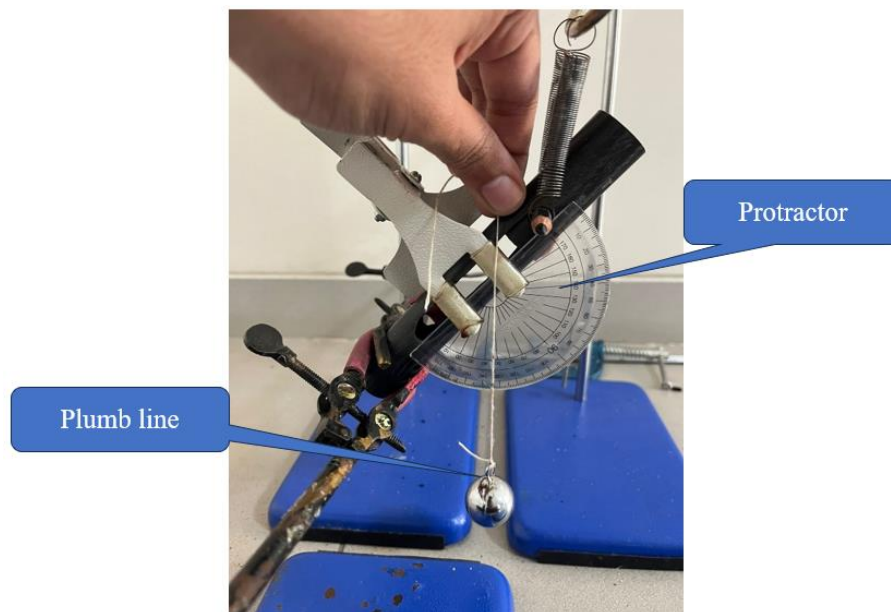
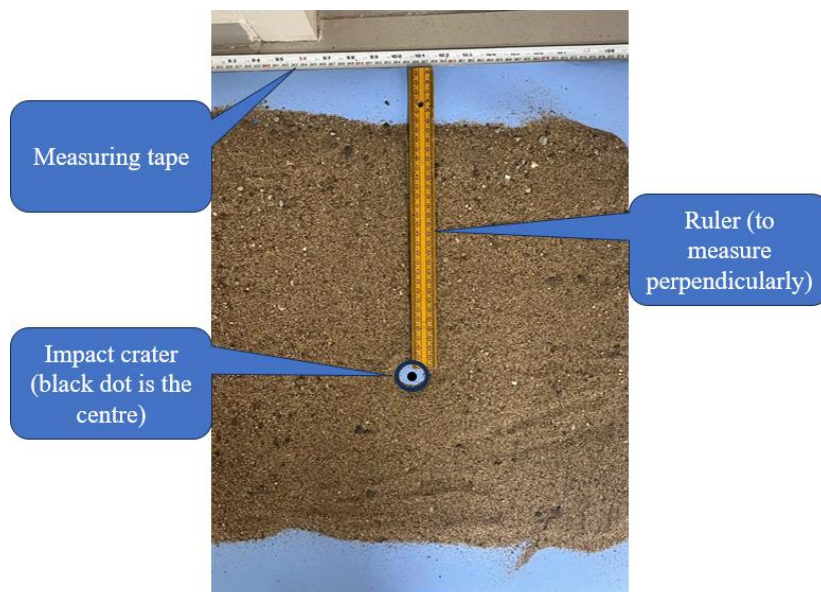


Figure 7: Range measurement



8. Safety, Ethical and Environmental Considerations

Safety Concerns:

1. Spring tension and release: The experiment uses tensioned springs, which can snap back violently, and even chip the apparatus. Thus, care was taken to handle and release the pusher carefully to avoid injury.
2. Risk of cuts: Cutting or modifying the PVC pipe can leave sharp edges, posing a risk of cuts. Proper protective gloves were used and edges were smoothed out using sandpaper.
3. Hot soldering iron usage: The hot soldering iron was operated in a well-ventilated area with heat-resistant gloves to avoid mishaps.
4. Unstable equipment: Improperly secured clamps or stands may fall and cause injury. All equipment was double-checked for stability.
5. Tripping hazards: The measuring tape and other equipment lying on the floor can create tripping hazards. The area was organized and caution signs were put up.

Ethical Considerations:

1. Misuse of experiment: The experiment shares similar working principles to weaponry such as cannons or guns. It is crucial to note that this experiment was strictly for educational purposes and fuelled by academic curiosity, with no other intention or underlying motive whatsoever.

Environmental Concerns:

1. Proper disposal of apparatus: Any materials, such as broken springs, PVC scraps or chips, sand and chart paper, etc. were disposed of in an environmentally friendly way.

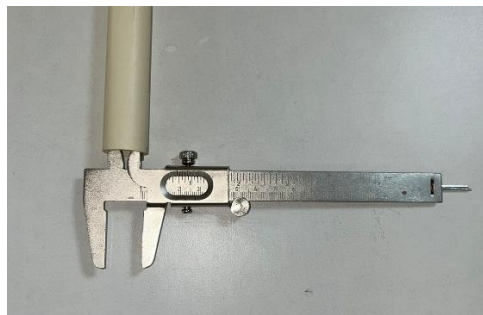
2. Energy consumption: Minimizing the use of electronic tools like the soldering iron can help reduce the overall carbon footprint of the experiment.

9.Data Collection and Analysis

Measurement of inner diameter of pipes:

The inner diameters of all pipes being used were measured using the internal jaws of a Vernier calliper as shown in Figure 8.

Figure 8: Measuring inner diameter of barrels



The diameters were recorded as in Table 5.

Table 5: Inner diameter of pipes

Inner diameter of barrel ($\times 10^{-2} \pm 0.000005$ m)
1.83
2.13
2.43
2.73
3.03
3.33

Measurement of projectile diameter:

Table 6: Raw data for marble diameter

	Trial 1	Trial 2	Trial 3
Diameter of projectile (marble) ($\times 10^{-2} \pm 0.000005$ m)	1.598	1.592	1.587

Then, the diameter of the marble was measured using a screw gauge with no zero error, as seen in Figure 9. To avoid random errors in measurement, 3 trials were taken, the average of which can be calculated as:

Figure 9: Measuring diameter of projectile



$$\text{Average diameter} = \frac{\text{Sum of all trials}}{\text{number of trials}} = \frac{1.598+1.592+1.587}{3} = 1.592 \times 10^{-2} \text{ m}$$

$$\text{Uncertainty} = 0.000005 \times 3 = \pm 0.000015 \text{ m}$$

Measurement of Windage:

An example calculation of windage:

$$\frac{\text{inner diameter}}{2} - \frac{\text{diameter of marble}}{2} = \frac{1.83}{2} - \frac{1.592}{2} = 0.119 \times 10^{-2} \text{ m}$$

Table 7: Windage

Inner diameter of barrel (x 10⁻² ± 0.00005 m)	Windage gap (x 10⁻² ± 0.00005 m)
1.83	0.119
2.13	0.269
2.43	0.419
2.73	0.569
3.03	0.719
3.33	0.869

$$\text{Uncertainty (UC)} = 0.00005 + 0.000015 = \pm \mathbf{0.000065 \text{ m}}$$

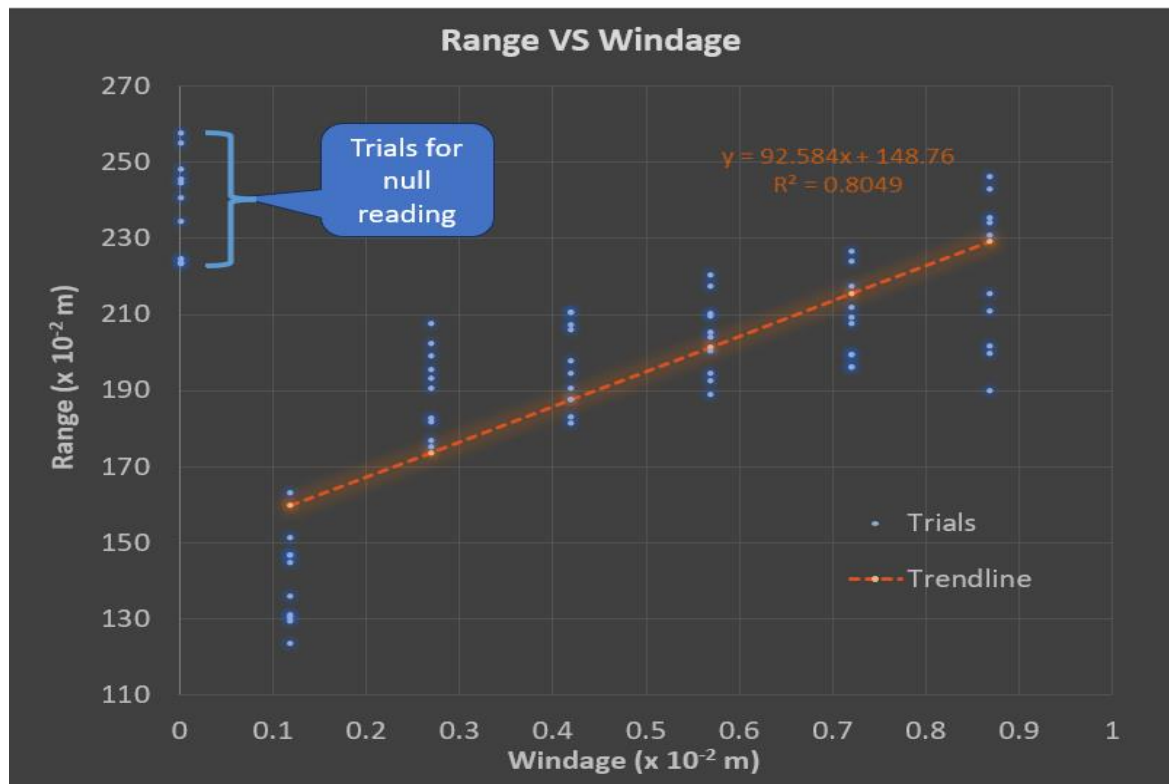
Next, the experiment was conducted following the procedure, and the results recorded are shown in Table 8 below.

Table 8: Raw Data table

Raw Data Table											
Inner diameter of barrel ($\times 10^{-2} \pm 0.00005$ m)	Windage gap ($\times 10^{-2} \pm 0.000065$ m)	Range ($\times 10^{-2} \pm 0.0005$ m)									
		T1	T2	T3	T4	T5	T6	T7	T8	T9	T10
-	(null reading)	248.3	257.5	224.4	244.5	255.2	223.1	223.5	240.7	245.6	234.4
1.83	0.119	151.3	135.9	129.5	144.6	163.2	146.8	123.4	146.6	130.4	131
2.13	0.269	177	202.4	207.3	182.9	190.3	175.2	181.8	195.4	199.1	193.2
2.43	0.419	190.4	207.1	181.4	206.1	183.1	210.4	187.5	194.4	197.8	210.3
2.73	0.569	194.3	200.1	220.3	204	209.9	205.1	192.4	209.6	217.1	189.1
3.03	0.719	196.1	199.3	223.8	226.5	209.2	217.1	207.4	211.9	196.3	199.4
3.33	0.869	190.1	242.8	201.5	230.8	210.7	233.9	245.8	235.6	199.7	215.3

And graphically these trials can be represented as:

Graph 1: Graphed data



(Note: The error bars are plotted but not visible as they are negligible as compared to the recorded data.)

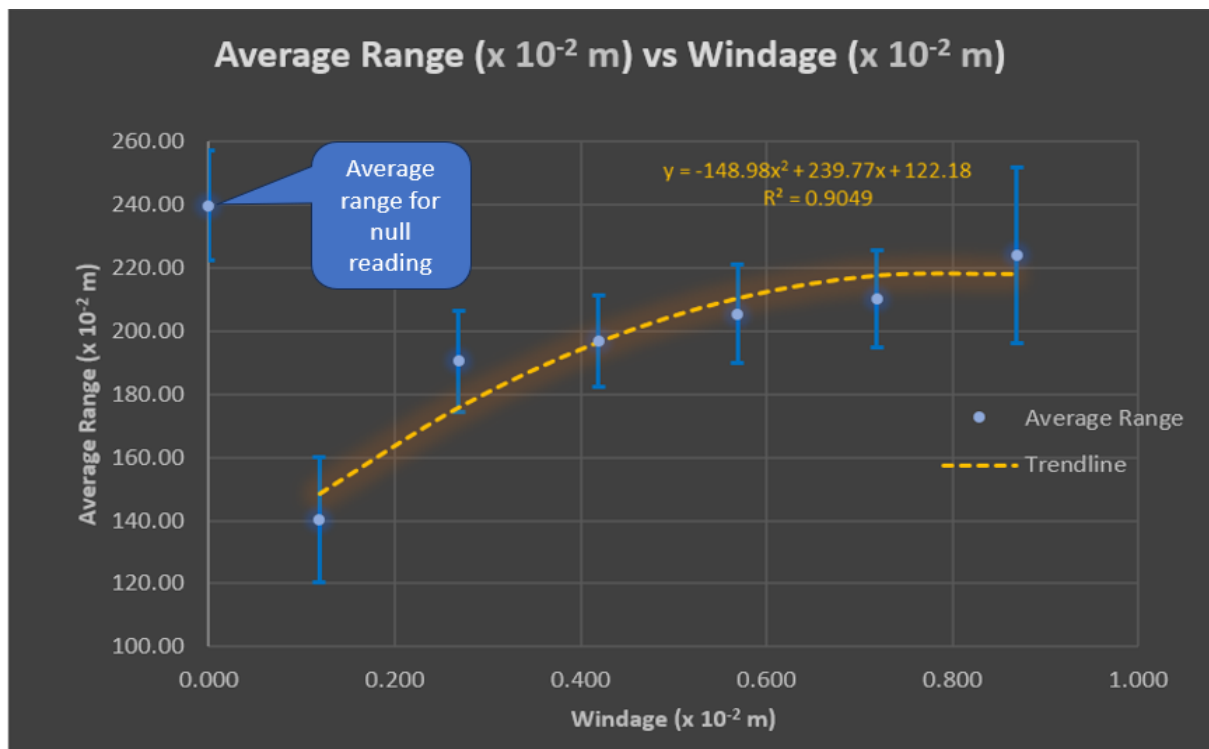
These values recorded over 10 trials are averaged out using Microsoft Excel to get an estimate of the range at each windage as follows, taking the null reading as an example:

$$\text{Average range} = \frac{\text{Sum of all trials}}{\text{number of trials}} = \frac{(248.3+257.5+\dots+234.4)}{10} = 239.72 \times 10^{-2} \text{ m}$$

Table 9: Processed data of Average range values with UC

Windage ($\times 10^{-2} \pm 0.000065$ m)	Average Range ($\times 10^{-2}$ m)
(null reading)	239.72 ± 17.2
0.119	140.27 ± 19.9
0.269	190.46 ± 16.05
0.419	196.85 ± 14.5
0.569	205.29 ± 15.6
0.719	210.10 ± 15.2
0.869	224.01 ± 27.85

Graph 2: Average range vs Windage



It can be observed that in Graph 1 that as windage increases, range also increases, but not proportionally. Graph 2 supplements this, showing a positive correlation between range

and windage. As the gradient changes as the windage increases, depicting a quadratic/polynomial trend of the 2nd order, which is seen in the trendline's equation in Graph 2. Additionally, the fact that the trendline passes through all the error bars and the R^2 value is very close to 1 show that this trendline is a good fit to the general trend. Initially, the gradient is steeper, but beyond 0.269×10^{-2} m windage, friction continues to decrease, but other factors, like air leakage and drag, start to dominate. The flatter gradient indicates that, after a point, the additional increase in windage has progressively less effect on the range. This suggests diminishing returns in range and possible stagnation beyond a threshold windage. This trendline contradicts null hypothesis, the reasons for which are explained later.

Furthermore, the distribution of data points for the X-value 0.869×10^{-2} m in Graph 1 is more than the others, suggesting a systematic error, maybe towards the end of the experiment the springs are deformed from use or have been stretched more than their point of elasticity. In Graph 2, the R^2 value is not that close to 1. This is because there is one anomaly at windage 0.119×10^{-2} m, which may be caused due to the larger effect of friction between barrel and projectile. The null reading was taken as a control since it is useful in providing the maximum value that range can reach as windage increases. Uncertainties and errors in experimentation have been explored in the Evaluation section.

Note- A small increment of $0.15 \times 10^{-2} \pm 0.000065$ m between each windage was used as a large increment may not show the actual effect of windage increase on range and also because the projectile diameter and inner diameter were already very small, thus to maintain relevance and proportionality, this small increment size was chosen. Also, 6 different radii were chosen since any less would not provide enough data to analyse and any more would not provide meaningful data since the windage would be too large to affect range as seen by the data trend becoming asymptotic as windage increases.

Theoretical Calculations of Range-

We can express range R in terms of time-of-flight T and launch angle θ , as follows:

Firstly, T can be expressed in terms of θ and initial velocity v_o as:

$$T = \frac{2v_o \sin \theta}{g}$$

Rearranging:

$$v_o = \frac{gT}{2 \sin \theta}$$

The horizontal component of velocity can be calculated as:

$$v_x = v_o \cos \theta$$

Substituting v_x into the Range equation:

$$R = v_x \cdot T = v_o \cos \theta \cdot T = v_o \cos \theta \cdot \frac{2v_o \sin \theta}{g} = \frac{2v_o^2 \sin \theta \cos \theta}{g}$$

Using the identity $2 \sin \theta \cos \theta = \sin 2\theta$ ("3.4: Projectile Motion"):

$$R = \frac{v_o^2 \sin (2\theta)}{g}$$

Substituting v_o :

$$R = \frac{\left(\frac{gT}{2 \sin \theta} \right)^2 \sin (2\theta)}{g}$$

Simplifying:

$$R = \frac{gT^2 \sin (2\theta)}{4 \sin^2 \theta} \quad (\text{Eq. 1})$$

Next, to calculate theoretical range using (Eq. 1), start and end times of the flight path were recorded via video analysis, as in

Table 10-

Table 10: Raw data values of start and end timings

		Raw Data for Time of Flight (± 0.01 s)						
Trial number	Windage gap ($\times 10^{-2}$ 0.000065 m) →	(null reading)	0.119	0.269	0.419	0.569	0.719	0.869
T1	Start	9.65	3.80	22.15	9.14	7.79	7.74	53.58
	End	10.59	4.52	22.90	9.91	8.60	8.58	54.44
T2	Start	4.70	9.28	32.79	5.80	8.77	7.91	10.93
	End	5.62	9.99	33.46	6.58	9.57	8.72	11.73
T3	Start	12.68	7.45	11.34	5.01	9.41	11.59	7.82
	End	13.65	8.14	12.13	5.74	10.20	12.41	8.66
T4	Start	6.79	5.76	7.75	3.39	8.31	6.73	14.21
	End	7.70	6.48	8.50	4.17	9.09	7.50	15.06
T5	Start	5.22	8.84	12.60	2.86	10.03	7.01	6.94
	End	6.14	9.50	13.38	3.62	10.83	7.81	7.73
T6	Start	6.40	8.16	23.29	3.79	6.67	7.63	22.58
	End	7.33	8.86	24.02	4.53	7.46	8.42	23.39
T7	Start	5.98	15.74	8.43	4.28	12.10	9.26	12.72
	End	6.89	16.40	9.14	5.07	12.87	10.00	13.52

T8	Start	7.42	5.28	14.02	3.71	9.15	7.38	6.82
	End	8.35	5.90	14.78	4.54	9.95	8.17	7.63
T9	Start	16.64	4.26	20.25	2.96	7.77	18.48	5.11
	End	17.61	5.00	21.02	3.70	8.55	19.31	5.97
T10	Start	5.58	5.67	7.98	3.05	7.91	6.83	7.04
	End	6.49	6.32	8.73	3.79	8.71	7.65	7.79

$$\text{Time of flight} = \text{End time} - \text{start time} = 10.59 - 9.65 = 0.94 \text{ s}$$

These values were averaged out as:

$$\text{Average time} = \frac{\text{Sum of all times}}{\text{number of trials}} = \frac{(0.94 + 0.92 + \dots + 0.91)}{10} = 0.931 \text{ s}$$

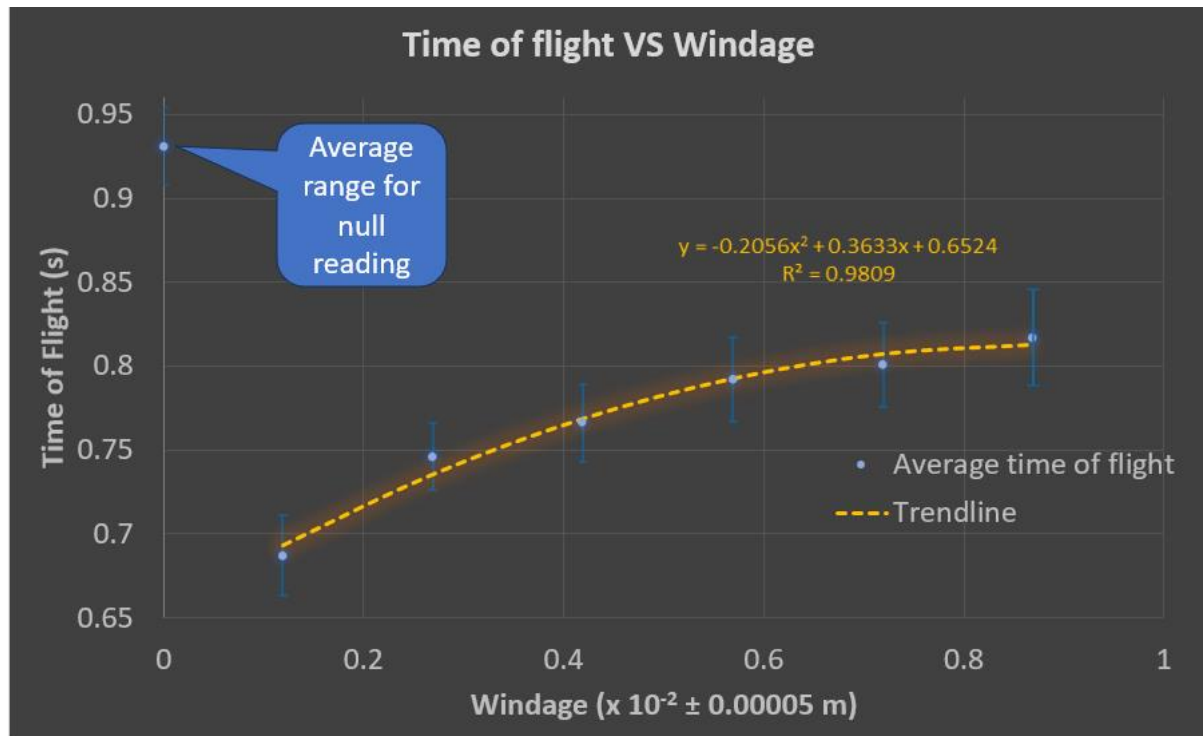
The processed data is shown in Table 11 below-

Table 11: Time of flight processed data

Windage gap (x 10 ⁻² ± 0.000065 m)	Time of Flight (± 0.01 s)										Average Time of flight (s)
	T1	T2	T3	T4	T5	T6	T7	T8	T9	T10	
(null reading)	0.94	0.92	0.97	0.91	0.92	0.93	0.91	0.93	0.97	0.91	0.931 ± 0.03
0.119	0.72	0.71	0.69	0.72	0.66	0.7	0.66	0.62	0.74	0.65	0.687 ± 0.06
0.269	0.75	0.67	0.79	0.75	0.78	0.73	0.71	0.76	0.77	0.75	0.746 ± 0.06
0.419	0.77	0.78	0.73	0.78	0.76	0.74	0.79	0.83	0.74	0.74	0.766 ± 0.05
0.569	0.81	0.8	0.79	0.78	0.80	0.79	0.77	0.8	0.78	0.80	0.792 ± 0.02
0.719	0.84	0.81	0.82	0.77	0.80	0.79	0.74	0.79	0.83	0.82	0.801 ± 0.05
0.869	0.86	0.80	0.84	0.85	0.79	0.81	0.8	0.81	0.86	0.75	0.817 ± 0.05

These can be graphed as:

Graph 3: Time vs Windage



Graph 3 can be seen to have a similar trend to Graphs 1 and 2, since range and time period are directly proportional. Using these values, theoretical values were calculated using (Eq.1) and compared to real values as in Table 12:

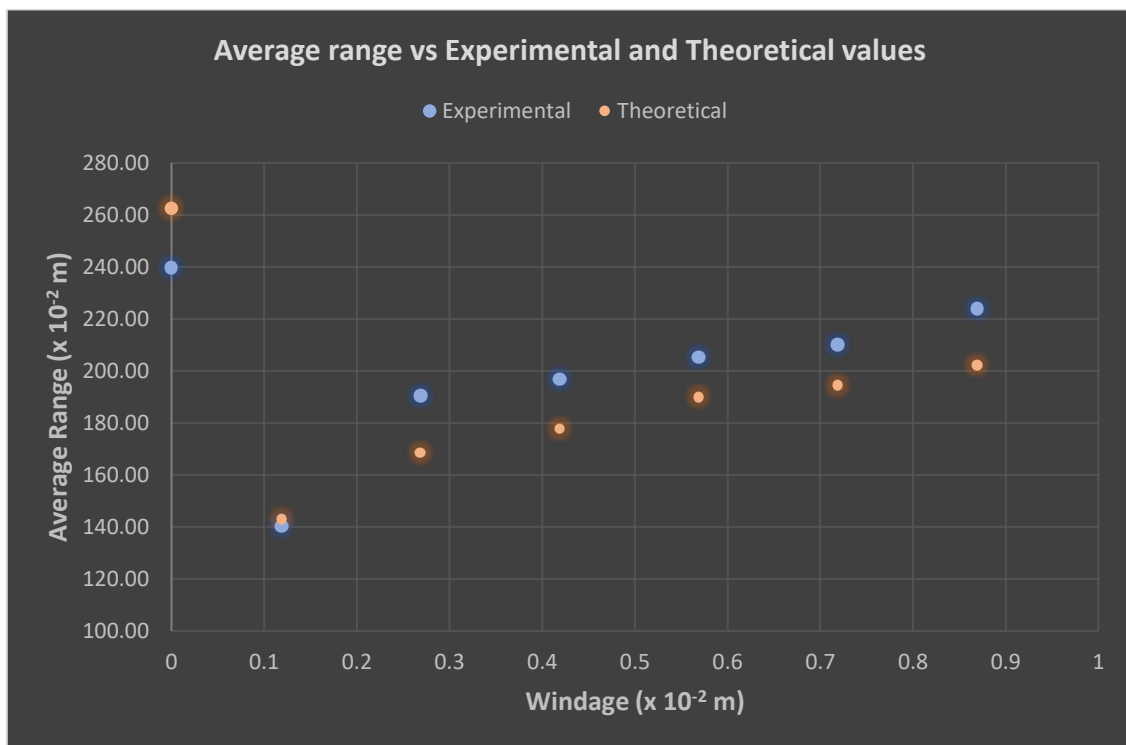
$$R = \frac{gT^2 \sin(2\theta)}{4 \sin^2 \theta} = \frac{9.81 \cdot 0.931 \cdot 0.931 \cdot \sin(45^\circ \cdot 2)}{4 \cdot \sin(45^\circ) \cdot \sin(45^\circ)} = 262.5 \times 10^{-2} \text{ m}$$

Table 12: Range comparison

Windage (x 10 ⁻² m ± 0.00005 m)	Average Time of flight (± 0.01s)	Experimental Range (x 10 ⁻² ± 0.0005 m)	Theoretical Range (x 10 ⁻² m)
(null reading)	0.931	239.72	262.5
0.119	0.687	140.27	142.9

0.269	0.746	190.46	168.5
0.419	0.766	196.85	177.7
0.569	0.792	205.29	189.9
0.719	0.801	210.10	194.3
0.869	0.817	224.01	202.1

Graph 4: Comparison between Experimental and Theoretical values of range



(Note: Since the theoretical values are calculated via (Eq.1), uncertainties of the apparatus are not taken into account)

In Graph 4, the experimental values are higher than theoretical, which may be due to various reasons:

- The timings were not precise enough as the video camera has least count of 0.01s.

- The frame rate of video camera might not allow to capture the exact moment when the pusher was left or the marble struck the ground, thus recorded values might be an overestimation of the actual time.
- Slight variations in tension or elasticity of springs used could have caused small fluctuations in initial velocity of the marble, influencing its range across trials, and hence time period.
- The theoretical time of flight assumes that projectile launches and lands at the same height. Although the setup was lowered, didn't touch the ground, and yet the measurement was taken where the marble hit the sand on the ground. This height difference between launch and landing point likely caused discrepancies between theoretical and experimental values.
- The exact moment the marble left the propulsion system wasn't clearly visible in the video recording, as its departure from the fully stretched spring lacked clarity. The lack of a distinct sound of release further complicated this, introducing uncertainty in start and end times of trials.

Error Percentage calculation:

$$\left| \frac{\text{Theoretical range} - \text{actual range}}{\text{theoretical range}} * 100 \right| = \left| \frac{262.5 - 239.72}{262.5} * 100 \right| = \mathbf{8.68\%}$$

Each error percentage is recorded below in Table 13.

Table 13: Error %

Windage ($\times 10^{-2}$ m $\pm$ 0.000065 m)	Error %
(null reading)	8.68
0.119	1.84
0.269	13.03
0.419	10.78
0.569	8.10
0.719	8.13
0.869	10.84

$$\text{Average Error \%} = \frac{(8.68+1.84+\dots+10.84)}{7} = \underline{\underline{8.77\%}}$$

10. Qualitative observation

Each time the marble was fired a small jerk-like movement in the barrel indicated contact. The most evident movement occurred with smallest diameter pipe, showing maximum contact when windage was least. This aligns with the data, as maximum friction with the smallest windage resulted in the least range. Friction decreased with increasing windage, reflected in the upward trend in the range shown in Graph 2.

Another observation was that the ball's horizontal path was deflected horizontally randomly. This could be due to the collisions that the ball had inside the barrel, thus changing its horizontal trajectory. However, this could also be due to incomplete or unequal loading of the pusher, thus introducing some degree of horizontal angle to the launching marble due to the angled force applied on it by the pusher.

11. Conclusion

After data collection and processing, it can be concluded that the null hypothesis that stated range would decrease as windage increases has been proven wrong by this experiment, due to several probable causes:

1. The hypothesis was made referring to projectiles such as bullets or cannon balls, which are high speed projectiles. This is a fault since this experiment uses spring mechanism which cannot produce such high speeds. Windage is important since it at high speeds, there is more effect of aerodynamic drag on the projectile than on lower speeds. Thus, due to difference in assumed speed in the hypothesis and experimentation, null hypothesis was proven wrong.
2. Initially, assumptions were made about the projectile assuming it will be propelled by compressed air, like hot air made from igniting gunpowder in cannons. However, the experiment uses a slingshot-like system to propel the marble, meaning that the propulsion is more guided as a singular force by the pusher acts on it in one direction, whereas with compressed air, the force acting on the marble would have been unpredictable and random, and it may also escape from the windage gap, thus increasing the importance and effect of windage on range.

Therefore, the experiment effectively rejects the null hypothesis and aligns with the alternate one for low-speed projectiles through fairly reliable methods as seen by low average error.

12.Evaluation

12.1 Weaknesses and Improvements

Source of error	Effect	Improvements
Systematic errors		
<i>Misalignment of launch angle</i>	Incorrect angles alter the trajectory, leading to significant deviation in range measurements.	Fix a 45° slant rule to the set-up for constant launch angle
<i>Friction in propulsion system</i>	Friction in the tube reduces launch speed, affecting consistency.	Regularly lubricate the system or use low-friction materials. Sanding barrel's inner surface will also help
<i>Propulsion system design</i>	The makeshift design of the set-up could lead to uneven launches.	Use a ready-made projectile launcher with an in-built protractor.
<i>Assumption of acceleration due to gravity</i>	Theoretical range calculations assume constant gravitational acceleration of 9.81 m s^{-2} , however due to air resistance, actual value may be less.	Use a heavier projectile to increase effect of gravity on it, and reduce effect of air resistance
<i>Deformation of springs</i>	Repeated use of springs could cause deformation, reducing force exerted and affecting range.	Ensure springs rest without extension when the system is not loaded. Replace springs if necessary.
<i>Timing device precision (0.01s camera)</i>	Lower precision timing might lead to misjudgement of flight time, impacting theoretical vs experimental data.	Use a higher frame rate camera or alternative high-precision timing

		methods for accurate time-of-flight measurements
Random Errors		
<i>Imprecise marking of marble's impact point</i>	There were errors in determining exact impact point due to the shifting sand used, which could result in inaccurate range measurements.	Cover the marble in washable paint and place a chart paper on the impact zone. This will give exact point where it struck the ground.
<i>Parallax error during measurement of range</i>	The range readings might have parallax error while recording data from the measuring tape	Use a digital measuring tape for better accuracy and removing parallax error

12.2. Extension for further research

1. The effect of varying launch angles on projectile range could provide insights into optimal launch conditions.
2. Analyse the impact of different projectile shapes and materials on aerodynamic drag and range could yield valuable data.
3. Examine how variations in spring stiffness and the design of the propulsion system influence energy transfer and projectile performance
4. These findings can be used to study propulsion systems that use spring-based systems such as ball-dispensing system used in sports, educational STEM kits, pin-ball machines, automated golf-putting machines etc.

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